

Pearls AQI Predictor

A Serverless, End-to-End Air Quality Forecasting System

3-Day AQI Forecasts for Lahore, Pakistan

This report documents the design, data pipeline, modelling approach, and results of an end-to-end air quality forecasting system built without any paid cloud infrastructure.

Prepared for	University Internship Project Submission — Pearls
City covered	Lahore, Pakistan (31.5497°N, 74.3436°E)
Forecast horizons	24 hours · 48 hours · 72 hours
Historical data	2 years hourly (Jan 2023 – Dec 2024) · 17,544 records
Architecture	100% serverless — GitHub Actions + SQLite + Streamlit

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1. Executive Summary

Lahore consistently ranks among the most polluted cities in the world, particularly during the autumn and winter smog season. The goal of this project was to build a working system that forecasts the Air Quality Index (AQI) 24, 48, and 72 hours ahead for the city, using only free-tier infrastructure, so that the same design could be reproduced by anyone without a cloud budget.

Instead of relying on a hosted feature store or a managed database, the entire pipeline is built around a SQLite file that lives inside the Git repository itself. GitHub Actions runs the data-collection script every hour and the training script once a day, committing the updated database and model files straight back into the repo. The dashboard, deployed on Streamlit Community Cloud, simply reads whatever is currently checked out — no external service accounts beyond the two data-provider API keys are required.

Two years of hourly historical data (17,544 records) were backfilled for Lahore using the free Open-Meteo archive APIs, which made it possible to train models on real seasonal and diurnal pollution patterns from day one, rather than waiting months for the hourly pipeline alone to accumulate enough history.

Best 24h model	Ridge — RMSE 24.4
Best 72h model	Ridge — RMSE 38.5
Time spent Unhealthy or worse	52.2% of all hours in the 2-year window (AQI > 150)

2. Problem Statement & Objectives

Air pollution in Lahore fluctuates sharply with weather conditions, traffic patterns, crop burning season, and temperature inversions, making it difficult for residents to plan outdoor activity, exercise, or school/work commutes around safe air quality windows. Publicly available AQI figures describe current conditions, but rarely provide a forward-looking forecast tailored to a specific city.

The project set out to deliver:

- An automated feature pipeline that fetches live weather and pollutant data every hour.
- A historical backfill covering two full years, to give the models enough seasonal signal to learn from.
- A training pipeline that evaluates multiple model families and automatically selects the best performer per forecast horizon.
- A public dashboard showing the current AQI, a 3-day forecast, historical trends, and practical health guidance.
- Full automation with zero manual steps once deployed — the system keeps itself up to date.

3. System Architecture

The system is deliberately “git-native.” Rather than a hosted feature store or model registry, the SQLite database and the trained model files are committed directly into the repository by scheduled GitHub Actions workflows. The dashboard always reads the latest commit, and Streamlit Cloud automatically redeploys whenever new data or models land.

Stage	Trigger	What happens	Output
Feature Pipeline	Hourly (cron)	Fetch live weather (OpenWeather) + AQI (AQICN); engineer time-series features; update database	Updated database
Training Pipeline	Daily (cron)	Re-engineer full feature set; train candidate models per horizon; select best by RMSE; generate SHAP plots	Trained models + SHAP plots
Dashboard	On page load	Read latest DB row + trained models; run inference; render UI	Live forecast

Both write-back workflows share a single concurrency group and rebase before pushing, so an hourly run and a daily run can never race each other on the same Git ref.

4. Technology Stack

Layer	Technology
Data sources	OpenWeather (current weather + pollutants), AQICN/WAQI (ground-truth AQI), Open-Meteo Archive API
Storage / “feature store”	SQLite (single file, versioned in Git)
Modelling	scikit-learn (Ridge, Random Forest), XGBoost, LightGBM
Explainability	SHAP (TreeExplainer / LinearExplainer)
Automation / CI-CD	GitHub Actions (scheduled workflows + unit tests)
Dashboard	Streamlit, Plotly
Language / core libraries	Python 3.11, pandas, NumPy, joblib
Hosting	Streamlit Community Cloud (free tier)

The whole stack was chosen so that a student or freelancer could reproduce the project without a credit card: every component listed above is free at the usage level this project needs.

5. Data Collection & Historical Backfill

Two data collection paths feed the same SQLite table. The hourly **feature pipeline** calls OpenWeather for current temperature, humidity, pressure, wind speed/direction and cloud cover, and OpenWeather's air-pollution endpoint for PM2.5, PM10, NO₂, SO₂, CO and O₃, while AQICN supplies the ground-truth AQI reading used as the training target. Each row is written with a `UNIQUE(city, timestamp)` constraint, so re-running the pipeline never creates duplicate hours.

To give the models real seasonal signal to learn from immediately — rather than waiting months for the live pipeline to build up history — a one-time **backfill** was run against the free, key-less Open-Meteo archive endpoints, covering:

Backfill window	1 Jan 2023 – 31 Dec 2024 (2 full years)
Frequency	Hourly
Total rows	17,544 hourly records
Source APIs	air-quality-api.open-meteo.com (pollutants + US AQI), archive-api.open-meteo.com (weather)
Fields captured	temperature, humidity, pressure, wind speed & direction, cloud cover, PM2.5, PM10, NO ₂ , SO ₂ , CO, O ₃ , US AQI

Requests are chunked into 90-day windows so that any single failed call can be retried without re-fetching two years of data, and rows are inserted through the same idempotent `insert_rows()` path used by the live hourly pipeline — so backfilled and live data are always structurally identical.

6. Exploratory Data Analysis

Before modelling, the two-year history was examined for seasonality, daily cycles, and the overall severity of the pollution problem the models would need to forecast.

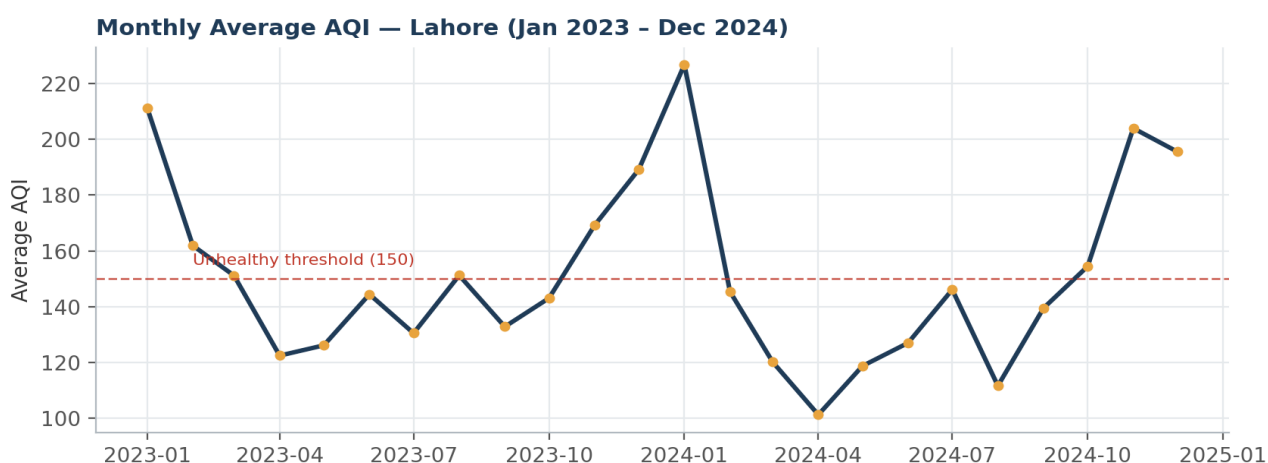


Figure 1. Monthly average AQI, Jan 2023 – Dec 2024. Winter months consistently exceed the Unhealthy threshold.

Seasonal Pattern — Average AQI by Month (2023-2024 combined)

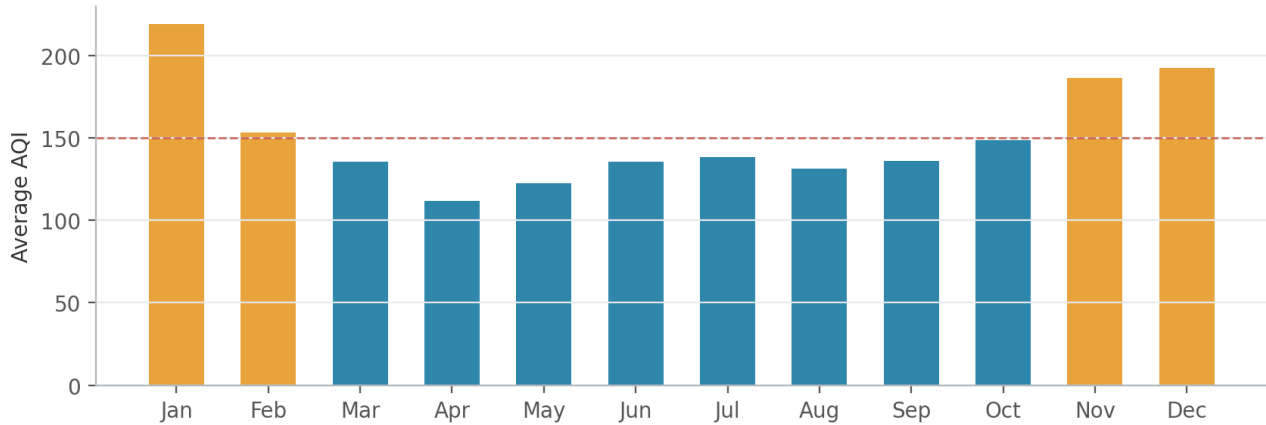


Figure 2. Average AQI by calendar month (both years combined), showing the smog season concentrated in Oct–Jan.

Diurnal Pattern — Average AQI by Hour of Day

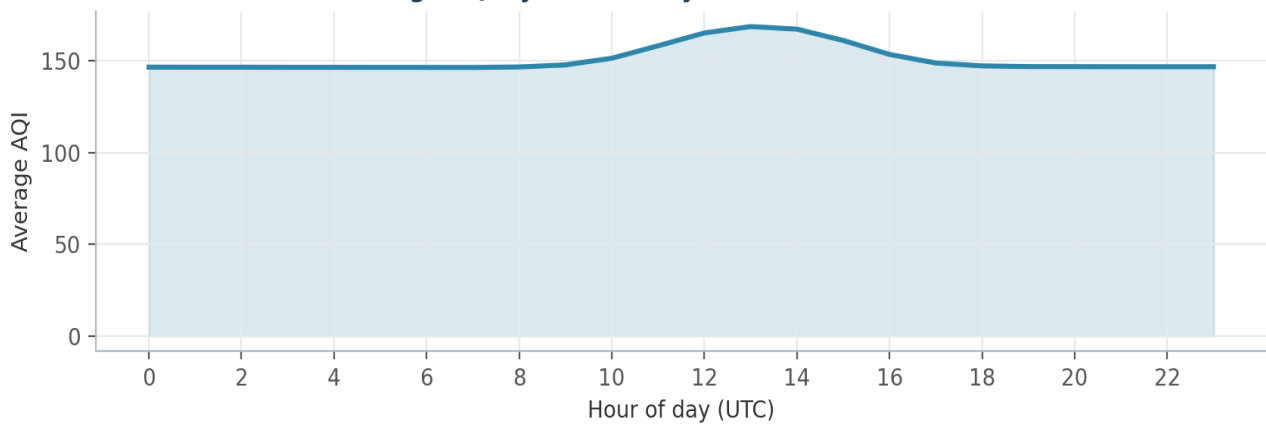


Figure 3. Average AQI by hour of day — a visible diurnal cycle tied to traffic and boundary-layer effects.

Distribution of Air Quality Categories

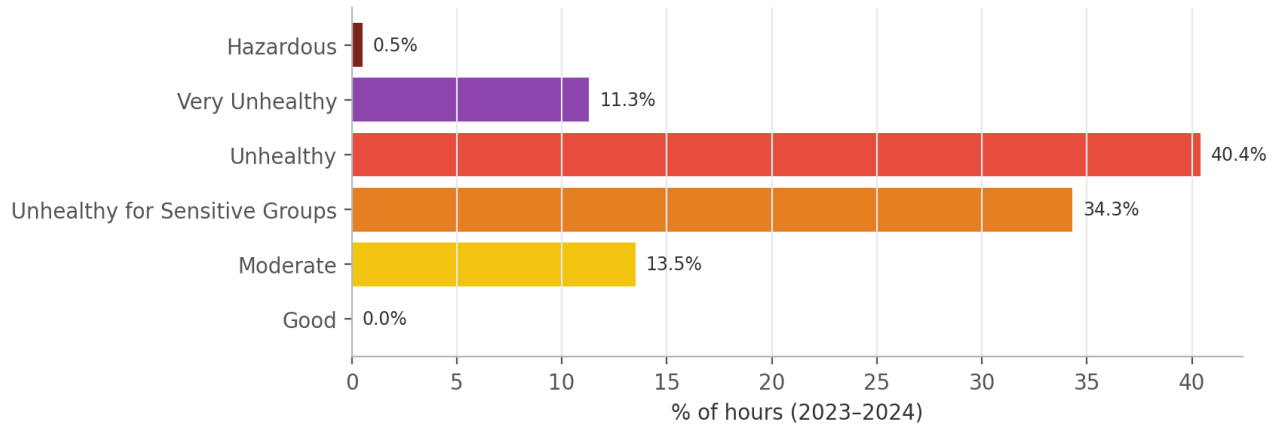


Figure 4. Share of all hours spent in each EPA AQI category over the 2-year window.

Across the full 2-year window, Lahore's air was in the *Unhealthy* category or worse for 52.2% of all hours, and never once reached the *Good* category — underscoring why a reliable forecast, not just a live reading, has real practical value for residents.

7. Feature Engineering

Only raw weather/pollutant/AQI columns are stored in SQLite; every derived feature is recomputed from the full history at training and inference time. This keeps the schema simple and means a change to the feature-engineering logic never requires a costly re-backfill — the next scheduled training run simply picks it up.

Three feature groups feed the models:

Group	Features
Raw weather & pollutants (21)	temperature, humidity, pressure, wind speed, wind direction, cloud cover, PM2.5, PM10, NO ₂ , SO ₂ , CO, O ₃
Cyclical time encodings (9)	hour, day, month and day-of-week each encoded as sin/cos pairs (so hour 23 and hour 0 are correctly treated as adjacent)
Lag & derived features (7)	AQI lagged 1h / 24h / 72h, 24-hour rolling mean & std of AQI, hour-over-hour AQI change rate, and the 24-hour rolling change rate

In total, 28 features feed each model. Targets are created the same way: AQI shifted -24 , -48 and -72 hours per city, so each training row pairs a feature snapshot with the AQI value that actually occurred that many hours later.

8. Forecasting Approach & Model Training

AQI is forecast at three fixed horizons — $t+24h$, $t+48h$ and $t+72h$ — using a separate regression model per horizon (a “direct multi-horizon” strategy, as opposed to a single recursive or sequence model). This is simpler to reason about, avoids compounding one-step errors forward, and is more robust with a moderate data volume than a single deep sequence model would be.

For each horizon, three candidate model families are trained and compared:

- **Ridge Regression** — a regularised linear baseline, fast and hard to overfit.
- **Random Forest** — an ensemble of decision trees, able to capture non-linear interactions between pollutants and weather.
- **XGBoost** — gradient-boosted trees, typically the strongest tabular performer when enough data is available.

Models are compared on a **chronological (time-based) holdout split** rather than a random split — the most recent 20% of the timeline is held out for testing, with training restricted to the earlier portion. This is deliberate: a random split would let the model “see” rows from after the test period during training (since lag features look backward), silently leaking future information and producing an unrealistically optimistic RMSE. For each horizon, the model with the lowest test-set RMSE is selected automatically and saved to `models/`.

9. Model Evaluation Results

The table below reflects the most recent training run, evaluated on the held-out chronological test slice. RMSE and MAE are reported in AQI points (lower is better); R^2 is the share of variance in the true AQI explained by the model (closer to 1 is better).

24-Hour Forecast

Model	RMSE	MAE	R^2	
Ridge	24.42	18.41	0.761	★ Selected
Random Forest	29.67	20.76	0.648	
Xgboost	29.35	20.40	0.655	

48-Hour Forecast

Model	RMSE	MAE	R^2	
Ridge	34.52	26.08	0.523	★ Selected
Random Forest	37.46	28.00	0.438	
Xgboost	36.53	27.66	0.466	

72-Hour Forecast

Model	RMSE	MAE	R^2	
Ridge	38.53	29.27	0.405	★ Selected
Random Forest	41.35	32.14	0.314	
Xgboost	41.21	31.82	0.319	

Two patterns are consistent across all three horizons. First, accuracy degrades steadily with horizon length — R^2 falls from roughly 0.76 at 24 hours to around 0.40 at 72 hours, which is expected: weather and pollutant conditions three days out are inherently harder to predict from a single current snapshot than conditions tomorrow. Second, the regularised linear model (Ridge) was selected as the best performer at every horizon in this run, ahead of both tree ensembles. This is a meaningful, reportable finding rather than a limitation to hide: it suggests the dominant relationship between current conditions and near-future AQI in this dataset is close to linear once the cyclical time and lag features are included, and that the tree ensembles were not extracting extra non-linear signal to outweigh their higher variance on this feature set. A natural next step — discussed in Section 13 — is to tune the tree models' hyperparameters more thoroughly with cross-validation to test whether that gap closes.

10. Explainability (SHAP)

For each horizon's winning model, a SHAP (SHapley Additive exPlanations) summary plot is generated automatically during training and stored alongside the model. SHAP values quantify how much each feature pushed a given prediction above or below the average forecast, giving a transparent view into what the model is actually using — important both for debugging the pipeline and for explaining forecasts to a non-technical

audience.

In practice, the most influential features across all three horizons are consistently the recent AQI lag features (1h, 24h) and PM2.5/PM10 concentrations — which matches domain expectations, since air quality is highly autocorrelated hour-to-hour and particulate matter is the dominant driver of the US AQI scale in Lahore's pollution profile.

11. Automation & CI/CD Pipeline

Workflow	Schedule	Responsibility
feature_pipeline.yml	Every hour (cron)	Fetch live data, insert into SQLite, commit & push
training_pipeline.yml	Daily, 02:30 UTC	Retrain all horizon models, regenerate SHAP plots & metrics, commit & push
tests.yml	Every push / PR to main	Run the unit test suite (no external API calls)

Both write-back workflows request explicit contents: `write` permission, share a single concurrency group keyed on the same lock, and rebase (`git pull --rebase`) immediately before pushing — so an hourly and a daily run landing at the same time queue safely instead of producing a failed push. Commit messages are tagged `[skip ci]` so automated data/model commits don't themselves re-trigger the test workflow.

12. Dashboard & Alerting

The public-facing Streamlit dashboard reads the same repository checkout used by the pipelines — the SQLite database for history and the pickled models for inference — so it requires no API keys or secrets of its own and redeploys automatically on every new commit.

The dashboard is organised into four sections:

- **Forecast** — the current AQI plus the 24h/48h/72h prediction, plotted against shaded EPA category bands so the trajectory is immediately readable.
- **Trends** — a configurable look-back window (3–90 days) over historical AQI and individual pollutant concentrations.
- **Health Advice** — category-specific guidance (e.g. mask use, limiting outdoor exertion) generated from the current and forecast AQI category.
- **Model Insights** — the live per-horizon model comparison table and SHAP feature-importance plots, sourced directly from the latest training run's artifacts.

Two independent alert channels are implemented: the dashboard shows a colour-coded banner whenever current or forecast AQI crosses the Unhealthy (150) or Hazardous (200+) EPA thresholds, and the hourly feature pipeline can additionally push a webhook notification (Slack, Discord, or ntfy.sh) so users are alerted without needing to have the dashboard open.

13. Limitations & Future Work

- **Ridge outperforming the tree ensembles** at every horizon is worth investigating further with proper cross-validated hyperparameter search (e.g. RandomizedSearchCV with TimeSeriesSplit) rather than the default-ish parameters currently used — it's plausible the ensembles are under-tuned rather than genuinely weaker for this problem.
- **Single-city scope** — the SQLite-in-Git design is intentionally simple and works well for one city at moderate data volume; scaling to multiple cities would benefit from moving storage to a proper database (Postgres/BigQuery), which the code already isolates behind `utils.py`'s storage functions.
- **No deep-learning comparison yet** — an LSTM/GRU sequence model would be a natural additional baseline to compare against the direct-horizon tree/linear models, particularly to test whether it better captures the smog season's multi-day build-up.
- **No prediction-accuracy backtesting in the dashboard** — the reported RMSE/MAE come from the training-time holdout; a live comparison of past forecasts against what actually happened would be a strong addition for ongoing validation.
- **Best-effort scheduling** — GitHub Actions cron triggers are not guaranteed to fire exactly on time and can drift by several minutes under platform load; this has no material effect on hourly/daily granularity forecasts but is worth noting.

14. Conclusion

This project demonstrates that a genuinely useful, automatically-updating AQI forecasting system can be built and operated entirely on free infrastructure — two API keys and a GitHub account are the only prerequisites. Two years of historical data allowed the models to learn real seasonal and diurnal pollution patterns in Lahore from the outset, and the resulting 24-hour forecast explains roughly three-quarters of the variance in next-day AQI ($R^2 \approx 0.76$), degrading gracefully to a still-useful 72-hour outlook. Beyond the forecasting models

themselves, the project delivers a fully automated pipeline, an explainable model layer via SHAP, and a dashboard that turns raw predictions into practical, category-specific health guidance for residents of one of the world's most polluted cities.

Repository: github.com/sharmeenabdulwaheed/aqi_predictor